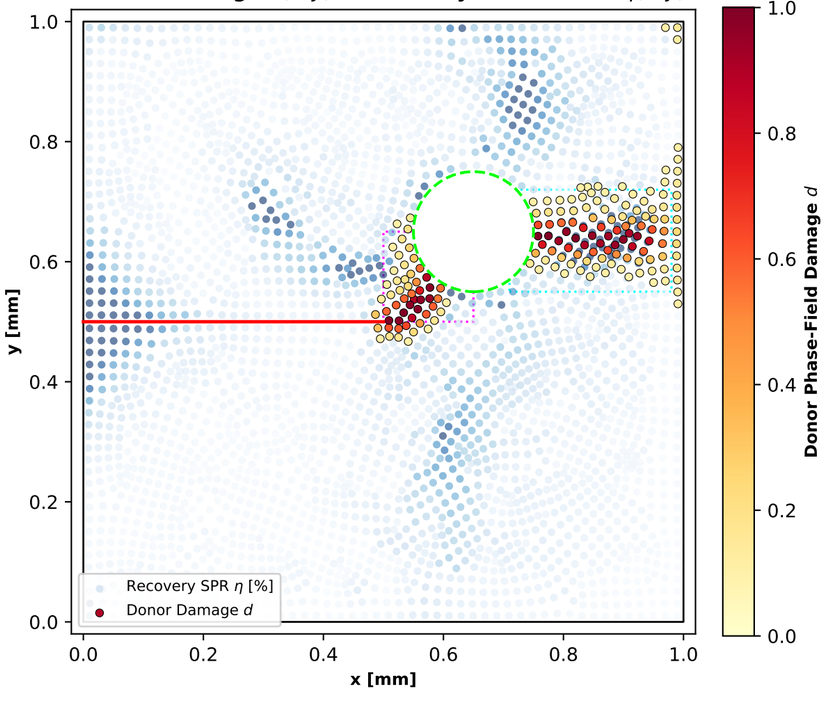
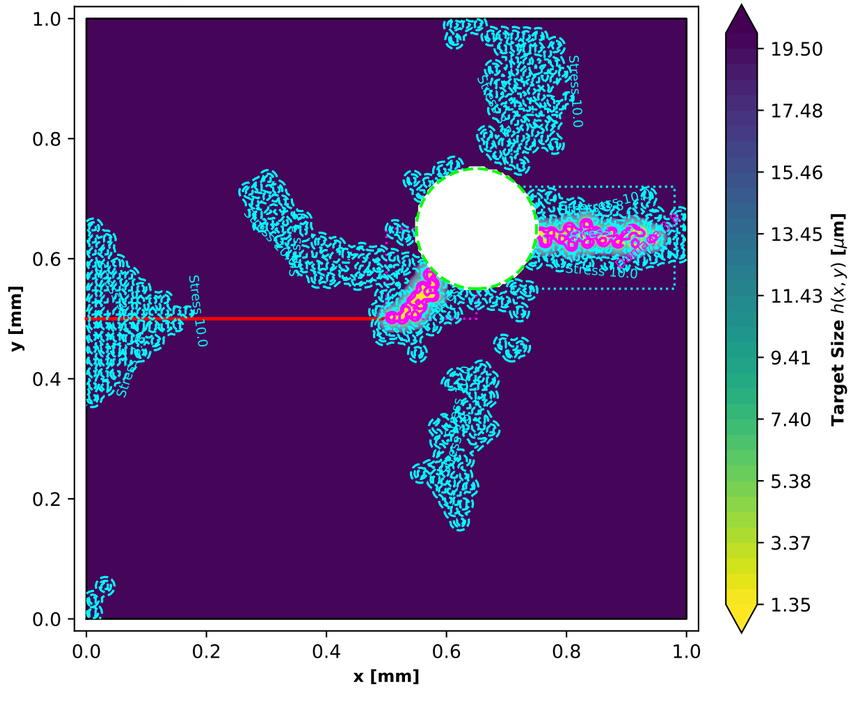
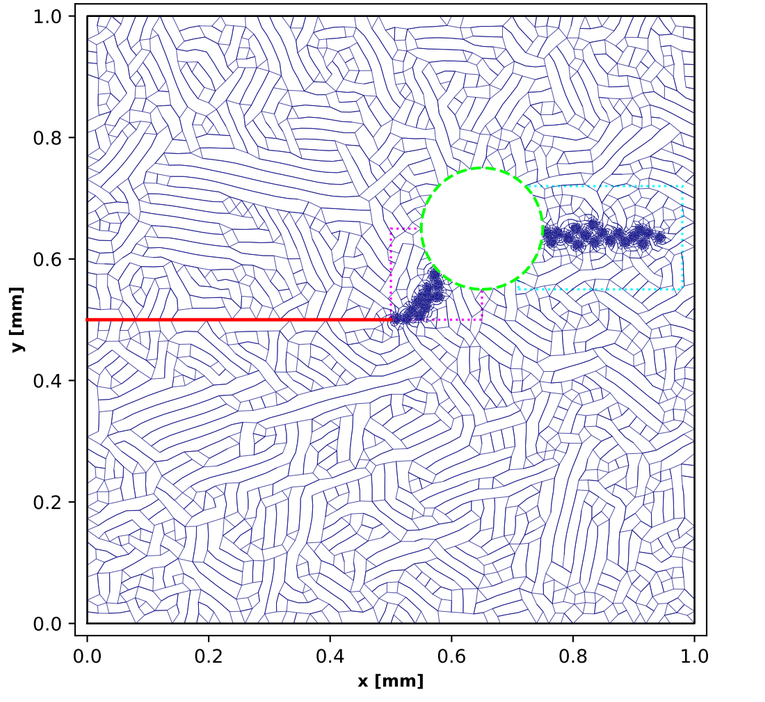
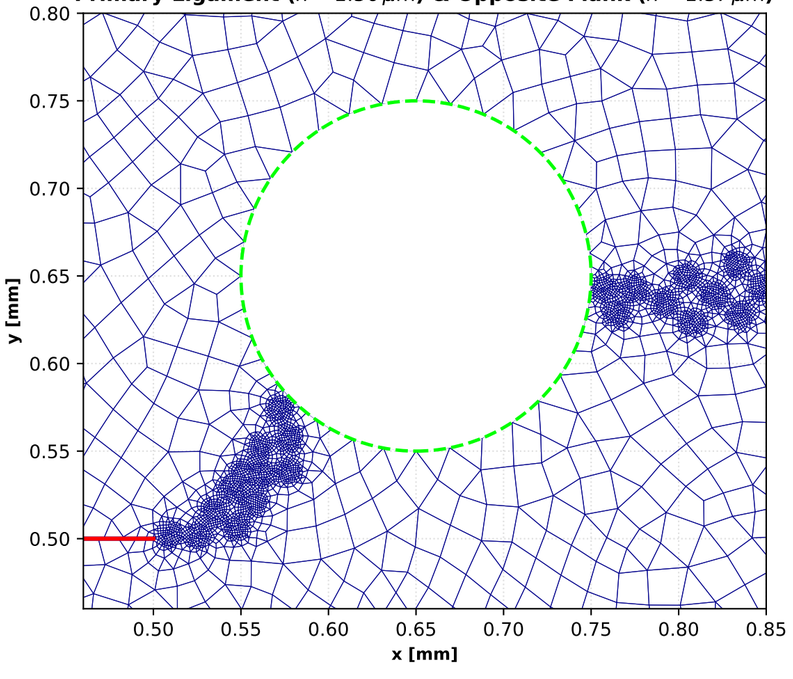
TASK F1668: CONTROLLED THREE-WAY ADAPTIVE REMESHING STUDY

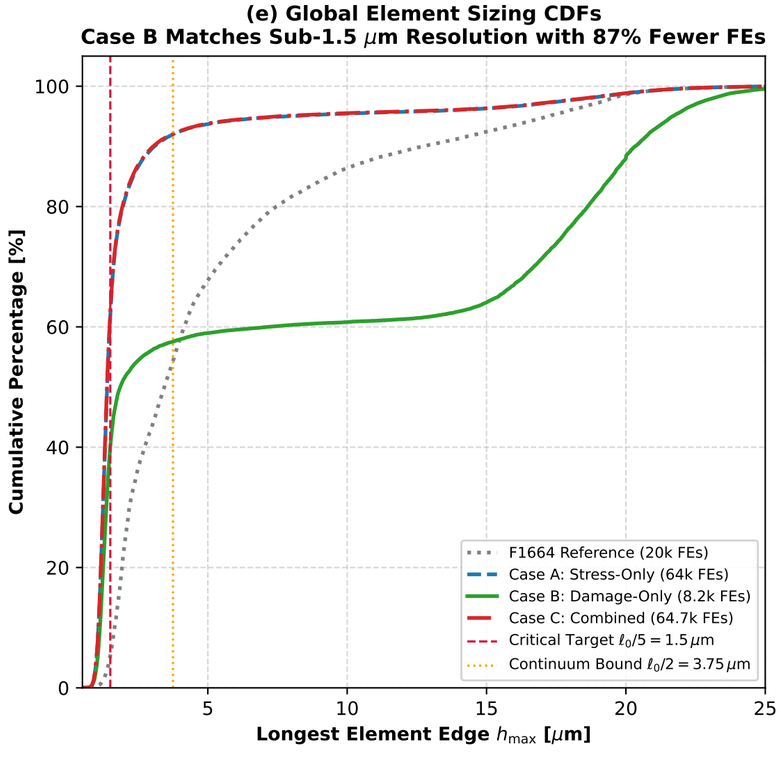
Mode-I Plate with Off-Axis Hole | Authentic Coarse Donor & Recovery Fields | Identical Geometry & Mesher











(f) Master Three-Way Scientific Evaluation Matrix
Case B Wins Efficiency-Resolution Trade-Off (8.2k vs 64.7k Quads)

Evaluation Metric	F1664 Ref	Case A (Stre	Case B (Dama	Case C (Hybrid
Total Elements	20,213	64,032	**8,237**	64,746
Pure Quads %	97.18%	78.03%	**77.67%**	78.09%
Non-Pos Jacobians	0	0	**0**	0
Min Signed det(J)	1.50e-7	1.10e-7	**1.08e-7**	1.29e-7
Sub-1.5 um Count	273 (1.4%)	27,554 (43%)	**2,299 (28%)**	28,097 (43%)
Ligament Median h	4.684 um	1.490 um	**1.501 um**	1.491 um
Advancing Tip Med h	2.091 um	1.532 um	**1.570 um**	1.516 um
Far-Field Median h	5.530 um	1.570 um	**20.609 um**	1.573 um
Far-Field Bloat %	Moderate	Severe (72%	**MINIMAL (<1%**	Severe (72%)
Efficiency Score	Moderate	Poor	**EXCELLENT**	Good-Accurate